4. DIVISION OF PARAMETERS (WHAT TO SIMULATE VS. WHAT TO TEST)

Parameters to Study via Simulation:

- **Wing Profile & Lift (Airfoil Selection):** Testing different wing shapes in software to find which generates the highest lift at low flying speeds.
- **Plane Stability & Balance (Center of Gravity):** Ensuring the wing and tail sizes are balanced so the plane does not flip or dive automatically in the air.
- **Wing Drag Optimization:** Adjusting the taper and twist of the wing in software to reduce air resistance.

Parameters to Verify Physically on the Ground/Field:

- **Actual Motor Thrust & Battery Drain:** Measuring how many grams of force the motor gives in real life and how fast the battery gets hot/drained.
- **Ground Resistance & Takeoff Roll:** Testing how smoothly the wheels roll on real grass or dirt during the takeoff run.
- **Wing Flexibility & Structural Strength:** Bending the physical wings by hand and during flight to make sure they do not snap under load.

5. VALIDATING SOFTWARE MODELS & PLANE DIAGNOSTIC PLAN

How to Validate Software with Real Data:
Measure the glide angle of the real plane during test flights. If the real plane drops faster than the software predicted, calculate the extra drag created by real-world gaps and tape, and feed that extra drag value back into the software model for future improvements.

Systematic Diagnostic Plan (If the Simulated Plane Cannot Take Off):
If a plane works in software but fails to rotate and take off on the field, follow this step-by-step diagnostic order:

1. **Check Wheels Position relative to Center of Gravity:** If the main wheels are placed too far behind the balance point, the tail control flap (elevator) cannot push the tail down to lift the nose up. *Fix: Move wheels slightly forward closer to the balance point.*
2. **Check Motor Power & Thrust:** Anchor the plane to a digital weighing scale and hit full throttle. If thrust is less than 50% of total plane weight, the motor/battery is weak or the propeller pitch is wrong. *Fix: Change battery or propeller size.*
3. **Check Wing Angle & Runway Friction:** If the runway grass creates too much wheel drag, or the wing is mounted completely flat, the plane cannot pick up speed. *Fix: Tilt the wing nose up by 2 degrees or use bigger wheels.*

6. FOUR-WEEK PROJECT SCHEDULE & CONTINGENCY TIME (BONUS)

- **Week 1 (Design & Software Testing):** Define mission requirements, select wing shapes in XFLR5, verify plane stability, and freeze the main plane dimensions.
- **Week 2 (CAD & Material Cutting):** Create 3D CAD models, test motor thrust on the bench scale, laser-cut plywood ribs, and hot-wire cut foam wings.
- **Week 3 (Assembly & Maiden Flight):** Glue parts together, install servos/receiver, balance the Center of Gravity, run taxi tests on the field, and perform the first maiden test flight.
- **Week 4 (Data Adjustment & Backup Days):** Tune the control flaps based on initial flight test results, re-verify with software, and reserve the last 3 days as contingency buffer for weather delays or crash repairs.